GeoAI-Based Wildfire Susceptibility Mapping and Community Risk Assessment in Missouri

UMSL Geospatial Collaborative

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WHY IT MATTERS

- Wildfire activity varies across Missouri.
- Identifying **high-risk areas and vulnerable communities** can support better preparedness and resource planning.

Wildfire Hazard → Susceptibility Mapping → Community Risk → Better Planning

OBJECTIVE

- Develop a GeoAI framework to map wildfire susceptibility across Missouri.
- Integrate historical wildfire occurrence with environmental and human factors.

① **Predict** elevated wildfire susceptibility
② **Identify** important environmental and human drivers
③ **Integrate** susceptibility with community exposure and vulnerability

DATA

DATA SOURCES

Multiple datasets were integrated to model wildfire susceptibility and assess community exposure and vulnerability across Missouri.

WILDFIRE OCCURRENCE
- **FPA-FOD** Wildfire occurrences (1992–2020)

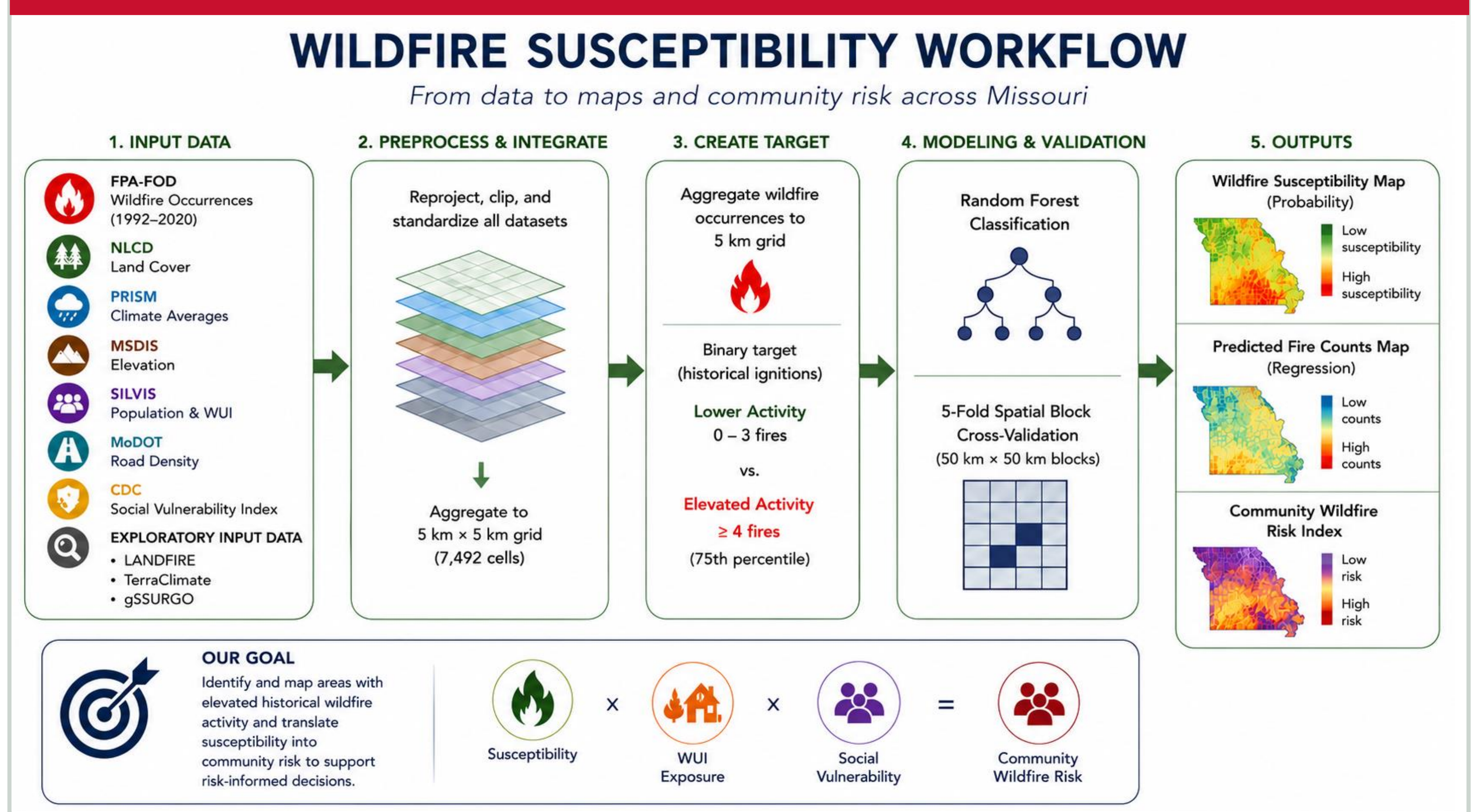
ENVIRONMENTAL & PHYSICAL
- **NLCD** Land cover (2021)
- **PRISM** Climate averages (1991–2020)
- **MSDIS** Elevation (1990)

HUMAN & INFRASTRUCTURE
- **SILVIS** Population & WUI exposure (1990–2020)
- **MoDOT** Roads & railroads
- **CDC SVI** Social Vulnerability Index (2022)

BOUNDARY
- **Missouri Department of Agriculture** MO Boundary

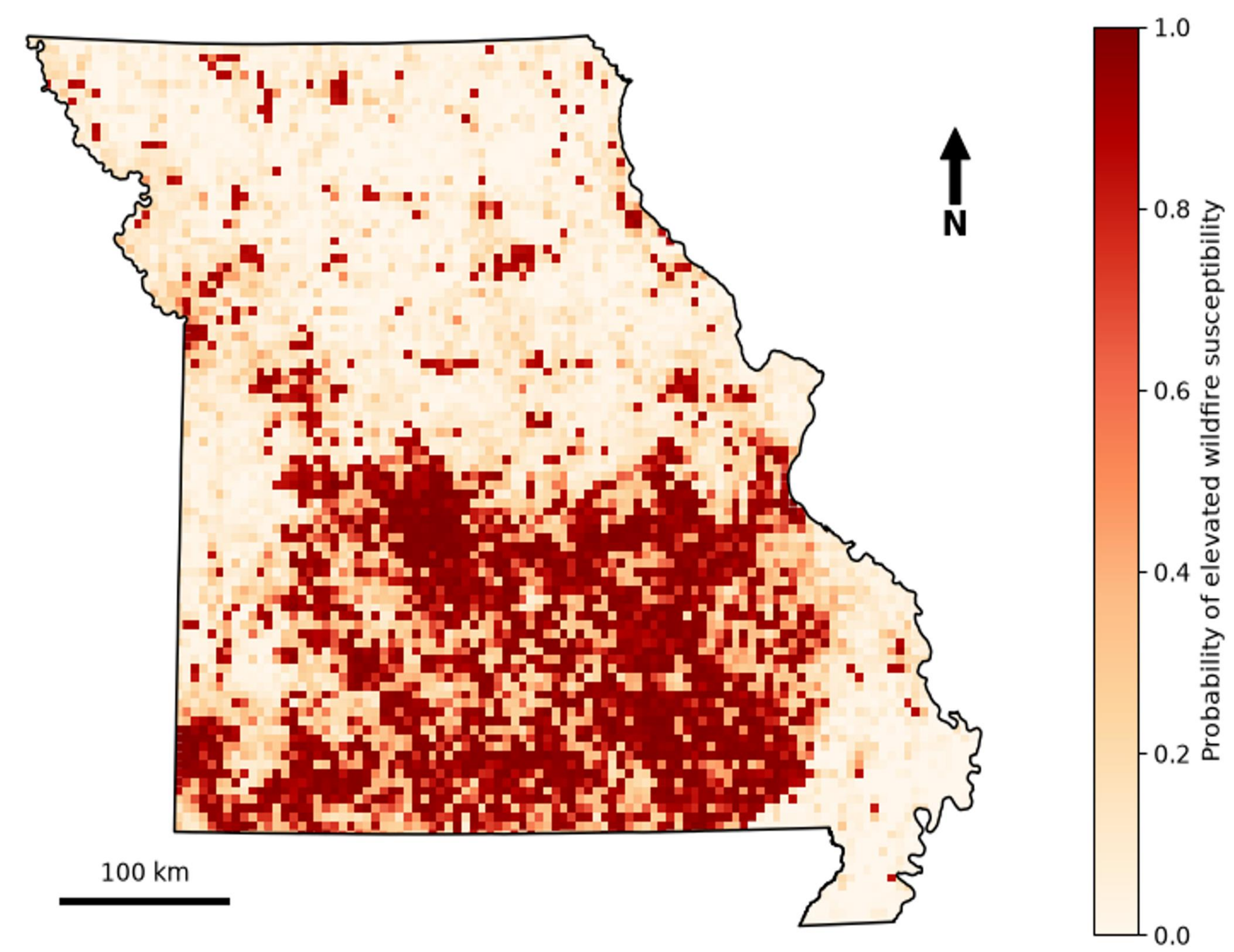
EXPLORATORY PREDICTORS
- **LANDFIRE** Fuel characteristics (2024)
- **gSSURGO** Soil properties (2025)
- **TerraClimate** Wind & drought variables (2015–2024)

WORKFLOW



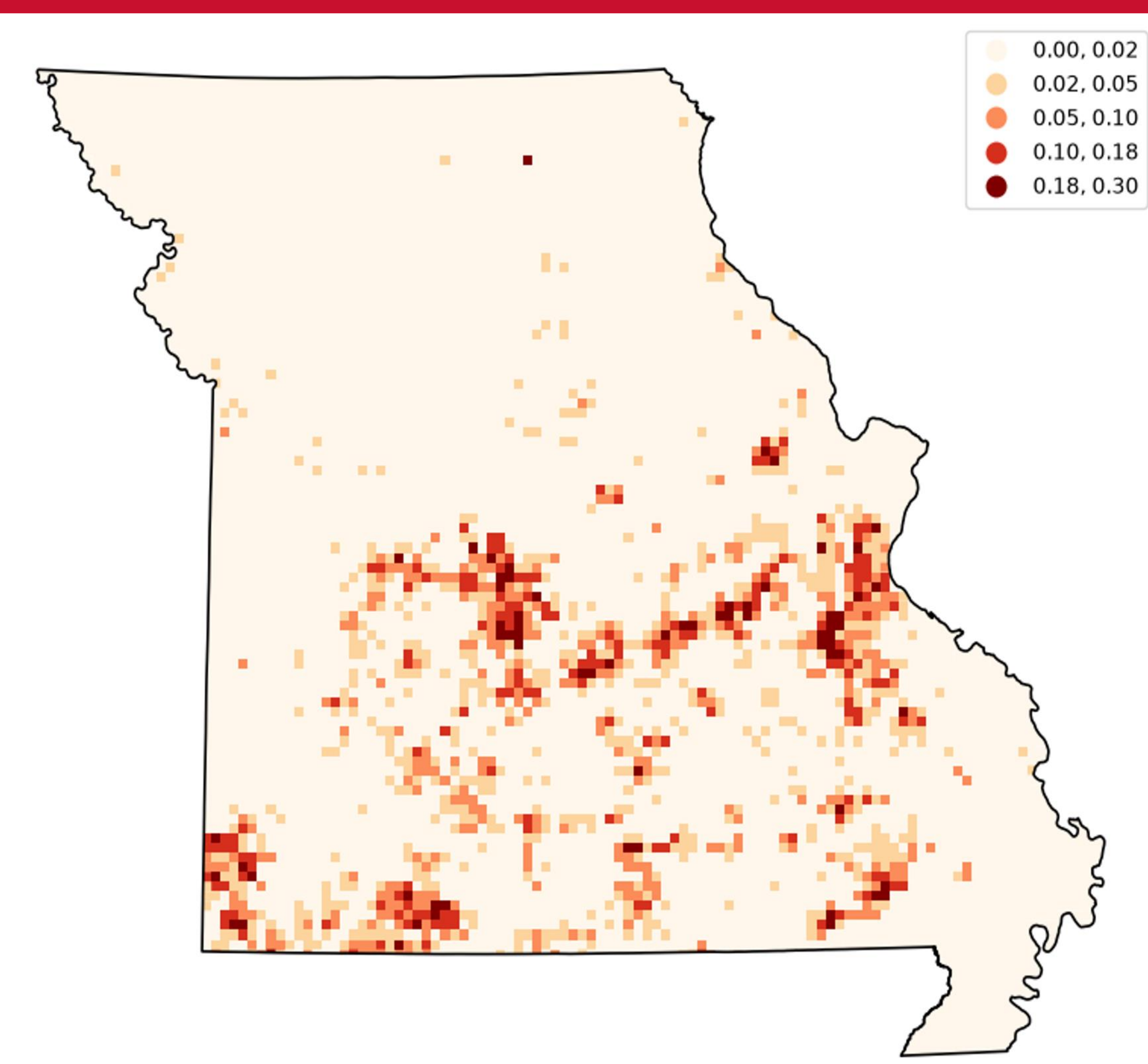
ArcGIS Pro + Python | Random Forest classification | XGBoost regression | 50 km spatial blocks

WILDFIRE SUSCEPTIBILITY



Random Forest probability of elevated historical wildfire activity across Missouri
Binary target: Elevated susceptibility = ≥4 historical fires per 5 km cell.

COMMUNITY WILDFIRE RISK



Communities where wildfire hazard, human exposure, and social vulnerability overlap
CRI = Susceptibility Probability × WUI Exposure × Social Vulnerability

MODEL PERFORMANCE

MODEL PERFORMANCE

Models were developed to predict wildfire susceptibility and estimate the number of historical fires across Missouri.

RANDOM FOREST CLASSIFICATION (Susceptibility)

Metric	Value	Description
ROC-AUC	0.88	Ability to distinguish between elevated vs. low susceptibility
Accuracy	81.5%	Overall prediction accuracy (random split)
Recall*	0.75	Identified elevated wildfire-activity cells
F1 Score*	0.70	Balance between precision and recall

*Elevated wildfire-activity class

SPATIAL VALIDATION

5-FOLD SPATIAL BLOCK CROSS-VALIDATION

ROC-AUC **0.83 – 0.87**

Predictive skill persisted across geographically separated test areas, demonstrating robust spatial generalization.

Spatial validation is crucial for geospatial models to ensure real-world transferability.

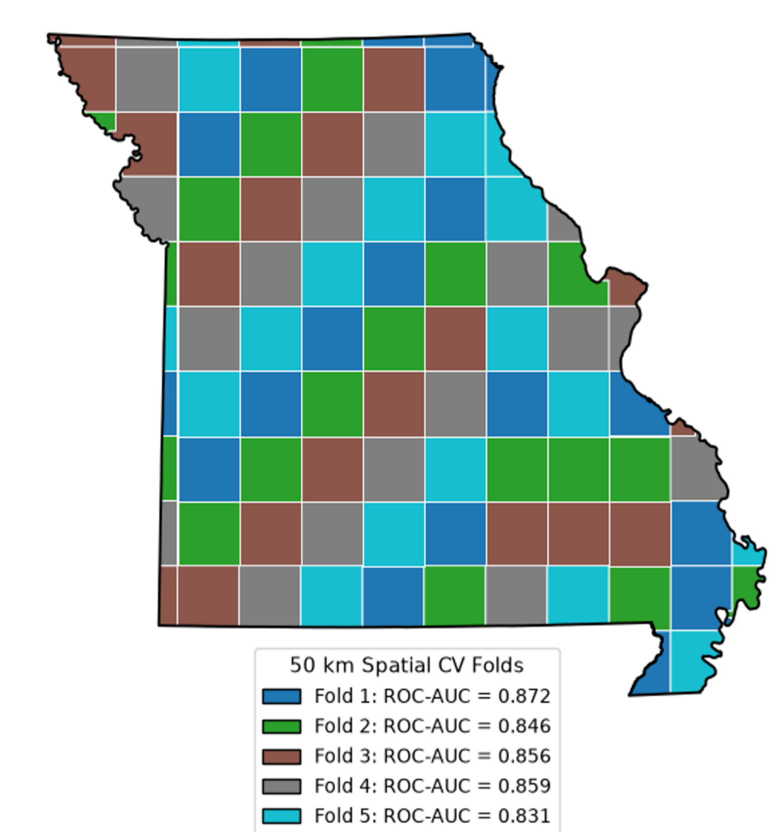
SECONDARY REGRESSION ANALYSIS: XGBoost captured broad spatial patterns but showed limited performance for extreme fire counts.

- MAE **2.33 fires** Mean Absolute Error
- RMSE **4.09 fires** Root Mean Squared Error

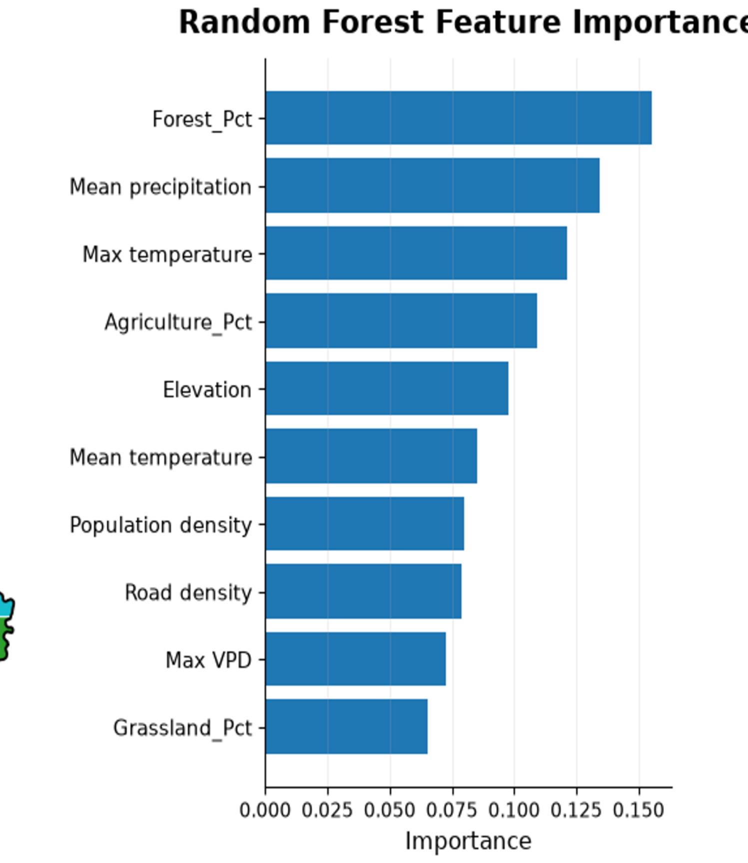
Exact fire-count prediction was challenging due to the strongly right-skewed distribution and underprediction of extreme values.

VALIDATION & EXPLANATION

Spatial Block Cross-Validation



Random Forest Feature Importance



KEY FINDINGS

High accuracy: 81.5%, AUC 0.88
Spatially reliable: AUC 0.83–0.87
Key drivers: Land cover, climate, terrain, and human factors
Better classification: Susceptibility predicted better than fire counts
Community risk: WUI and vulnerability identified higher-risk areas

References

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